

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Viva Voce Questions

Course Code:	DSA0405	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO4 – Viva Voce Question	Assessment:	Assessment Tool 3– Viva Voce Question
Weightage:	40% of CIA	Total Marks:	100
CO4 Statement:	Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BL4)		
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Section / Batch:	2024- CSE(AI)	Date:	20-08-26

Code of Conduct

I __ S.Sirivally / 192472371 __ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: __ Sirivally.S __

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	

Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

A. Introduction to Statistical Inference

1. What is statistical inference?

Statistical inference is the process of using sample data to make conclusions or predictions about a larger population.

2. Why is statistical inference important in data science?

It helps data scientists make decisions about a population using limited sample data. It is useful for prediction, estimation, and hypothesis testing.

3. What is the difference between descriptive statistics and inferential statistics?

Descriptive statistics summarize the data we have, using measures such as mean, median, and standard deviation. Inferential statistics use sample data to make conclusions about a population.

4. What is a population in statistics?

A population is the complete set of individuals, objects, or observations that we are interested in studying.

5. What is a sample?

A sample is a smaller subset selected from the population for analysis.

6. Why do data scientists usually work with samples instead of complete populations?

Studying the entire population may be expensive, time-consuming, or impossible. A properly selected sample can provide useful information about the population.

7. What is a population parameter?

A population parameter is a numerical value that describes a population. Examples are population mean μ and population proportion p .

8. What is a sample statistic?

A sample statistic is a numerical value calculated from sample data. Examples are sample mean $\bar{x}$ and sample proportion $\hat{p}$.

9. Give a real-world example where statistical inference is used in industry.

A company may survey 1,000 customers and use their responses to estimate the satisfaction level of all customers.

10. What are the two major activities involved in statistical inference?

The two major activities are **estimation** and **hypothesis testing**.

B. Frequentist Approach

11. What is the frequentist approach to statistical inference?

The frequentist approach uses repeated-sampling ideas to make conclusions from data. Population parameters are treated as fixed but unknown quantities.

12. What is the basic idea behind frequentist statistics?

The basic idea is to use observed sample data and understand how results would behave over many repeated samples.

13. How does the frequentist approach differ from the Bayesian approach?

Frequentist statistics treats parameters as fixed unknown values. Bayesian statistics treats parameters as uncertain quantities and incorporates prior information.

14. In the frequentist approach, what is considered fixed: the parameter or the data?

The population parameter is considered **fixed**, while the sample data are considered **random** because different samples can produce different observations.

15. What does long-run frequency mean in frequentist statistics?

It means the proportion of times an event occurs when an experiment is repeated many times under the same conditions.

16. Why is random sampling important in the frequentist approach?

Random sampling helps ensure that the sample represents the population and allows probability-based statistical inference.

17. Can a population parameter be considered a random variable in the frequentist approach? Why?

No. In the frequentist approach, the parameter is treated as a fixed but unknown value. The randomness comes from the sample.

18. Give an example of a frequentist problem in a data science application.

A data scientist can test whether a new machine-learning model has significantly higher accuracy than an existing model using a random test sample.

C. Point Estimation

19. What is a point estimate?

A point estimate is a single numerical value used to estimate an unknown population parameter. For example, the sample mean estimates the population mean.

20. What is an estimator?

An estimator is a rule or formula used to calculate an estimate of a population parameter from sample data.

21. What is the difference between an estimator and an estimate?

An **estimator** is the formula or method, while an **estimate** is the numerical result obtained after applying that formula to a particular sample.

22. What is the point estimator for the population mean?

The sample mean, $\bar{x}$, is the point estimator for the population mean μ .

23. How do you estimate a population proportion from a sample?

We use the sample proportion:

$$\hat{p} = \text{Number of successes} / \text{Sample size}$$

24. What is the point estimate of population variance?

The sample variance, s^2 , is commonly used as the unbiased estimator of population variance σ^2 .

25. Why is the sample mean commonly used to estimate the population mean?

The sample mean is easy to calculate and, under standard random-sampling assumptions, is an unbiased estimator of the population mean.

26. What properties should a good estimator have?

A good estimator should preferably be **unbiased, consistent, and efficient**.

27. What is an unbiased estimator?

An estimator is unbiased if its expected value equals the true population parameter.

For example:

$$E(\bar{x}) = \mu$$

28. What is meant by the bias of an estimator?

Bias is the difference between the expected value of an estimator and the true population parameter.

29. What is the difference between bias and variance of an estimator?

Bias measures systematic error from the true parameter. **Variance** measures how much the estimator changes from sample to sample.

30. Why can two different samples produce different point estimates for the same population parameter?

Because samples contain different observations. Therefore, statistics such as the sample mean can vary from one sample to another.

D. Variability in Estimates

31. Why does a point estimate have uncertainty?

A point estimate is calculated from a sample rather than the entire population. Different samples can give different estimates.

32. What is the sampling distribution of an estimator?

It is the probability distribution of an estimator obtained from all possible samples of the same size from a population.

33. What is the standard error?

Standard error measures the variability of a sample statistic across repeated samples. For the sample mean, it is commonly:

$$SE = \sigma / \sqrt{n}$$

or estimated using $s / \sqrt{n}$.

34. How is standard error different from standard deviation?

Standard deviation measures variability among individual observations. Standard error measures variability of a statistic, such as the sample mean.

35. How does increasing the sample size affect the standard error of the sample mean?

Increasing the sample size decreases the standard error.

Because:

$$SE = \sigma / \sqrt{n}$$

36. Why does variability in an estimate decrease when the sample size increases?

A larger sample contains more information about the population, so random sampling fluctuations have less influence on the estimate.

37. What is the relationship between standard error and confidence intervals?

Standard error determines how much uncertainty is associated with an estimate. A larger standard error generally produces a wider confidence interval.

38. If two samples have different sample means, does it necessarily mean that one sample is incorrect? Explain.

No. Different random samples naturally produce different sample means. This variation is called sampling variability.

E. Confidence Intervals

39. What is a confidence interval?

A confidence interval is a range of values calculated from sample data that is used to estimate an unknown population parameter.

40. Why is a confidence interval more informative than a point estimate?

A point estimate gives only one value, while a confidence interval gives a range and indicates the uncertainty associated with the estimate.

41. What does a 95% confidence level mean in the frequentist approach?

If we repeatedly take samples and construct confidence intervals using the same method, approximately 95% of those intervals would contain the true population parameter.

Important viva point:

It is not technically correct to say that there is a 95% probability that the fixed parameter is inside this particular interval.

42. What is the difference between a confidence level and a confidence interval?

The **confidence level** is the reliability level of the method, such as 95%. The confidence interval is the actual numerical range obtained from the sample.

43. What factors determine the width of a confidence interval?

Main factors are:

- Sample size
- Variability or standard error
- Confidence level

44. How does increasing the sample size affect the width of a confidence interval?

Increasing the sample size generally makes the confidence interval **narrower** because the standard error decreases.

45. How does increasing the confidence level from 95% to 99% affect the confidence interval?

The interval becomes **wider** because we need a larger range to achieve greater confidence.

46. What is the difference between a z-confidence interval and a t-confidence interval?

A z-confidence interval uses the standard normal distribution, commonly when the population

standard deviation is known or large-sample conditions justify it. A t-confidence interval uses the t-distribution, especially when the population standard deviation is unknown and the sample is relatively small.

47. When would you use the t-distribution instead of the normal distribution for estimating a population mean?

When the population standard deviation is unknown and is estimated using the sample standard deviation, particularly for small samples, we generally use the t-distribution.

F. Hypothesis Testing Using Confidence Intervals

48. What is hypothesis testing?

Hypothesis testing is a statistical method used to determine whether sample data provide sufficient evidence against a claim about a population.

49. What are the null hypothesis (H_0) and alternative hypothesis (H_1)?

The null hypothesis (H_0) represents the default claim or no significant effect. The alternative hypothesis (H_1) represents the effect, difference, or relationship that we want to investigate.

50. How can a confidence interval be used to make a decision about a hypothesis?

We compare the hypothesized parameter value with the confidence interval.

For example, if the hypothesized value is $\mu = 50$ and the 95% confidence interval is (52, 58), then 50 is outside the interval, so we would reject H_0 at the corresponding 5% significance level for a two-sided test.